CHAPTER 4.1

The Impact of Stress on Serotonergic Neurotransmission

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Abstract: Levels of stress perceived by people in our modern society seem to have risen considerably, a phenomenon which may be related to the increased incidence of stress-related mental health problems such as depression and anxiety disorders. Whereas the body is well equipped to deal with single, acute stressful challenges, regulatory systems may derail when stress is of a repeated or prolonged nature. Two of the systems that appear to be malfunctioning in psychiatric disease are the hypothalamic–pituitary–adrenocortical (HPA) axis (i.e., the stress hormone axis) and the serotonin (5-HT) system. This chapter will focus on the effects of stress on serotonergic neurotransmission. We will first discuss how stress affects the activity of the serotonergic cell bodies in the rostral (midbrain/pontine) raphe nuclei (the dorsal and median raphe nucleus), followed by an overview of the effects of stress on 5-HT synthesis. The dorsal and median raphe nuclei project to major limbic forebrain structures involved in the coordination of the stress response, such as the frontal cortex, hippocampus, amygdala and hypothalamus. Therefore, we will subsequently highlight the effects of stress on the extracellular levels of 5-HT in these structures as assessed by *in vivo* microdialysis. Next, the (aberrant) functioning of serotonergic neurotransmission in animal models with changes in HPA axis regulation will be discussed, followed by an excursion into the beneficial effects of moderate exercise on stress coping mechanisms.

The reviewed data clearly demonstrate that the neurotransmitter 5-HT is not only highly responsive to acute stress; its responses are also intricately coordinated in a stressor-dependent and brain region-specific manner. This notion holds true for the regulation of the firing rate of serotonergic neurons, the activity and expression of tryptophan hydroxylase (the rate-limiting enzyme in the synthesis of 5-HT) and the extracellular levels of 5-HT in terminal regions. Furthermore, these data show that the 5-HT system can become compromised during situations of aberrant functioning of the HPA axis – for example, due to changes in the corticotropin-releasing factor system or in corticosteroid receptor function. Most data presented here are obtained during and following acute stressful challenges. This information is of great relevance for our understanding of the relationship between stress and 5-HT. However, there is unfortunately only very limited information available on the effects of repeated and chronic stress. With respect to mental health disease (and animal welfare), it will therefore be essential to make the next step and increase our knowledge on how prolonged stress may perturb serotonergic neurotransmission. Finally, as stress will not suddenly 'disappear' from our society, and the costs of stress-related disease are burdening the health systems of many countries, it is important to study the mechanisms underlying adequate coping with stress and stress resilience, and to find ways to improve and support such mechanisms – for example, by exercise.

Abbreviations: 5-HT, serotonin (5-hydroxytryptamine); 5-HIAA, 5-hydroxyindoleacetic acid; CRF, corticotropin-releasing factor; CRF1, corticotropin-releasing factor receptor type 1; CRF2, corticotropin-releasing factor receptor type 2; DRN, dorsal raphe nucleus; GR, glucocorticoid receptor; HPA, hypothalamic-pituitary-adrenocortical; MRN, median raphe nucleus; SSRI, selective serotonin reuptake inhibitor; TPH, tryptophan hydroxylase.

Introduction

In our modern society stress seems to be an omnipresent phenomenon. Many people complain about having to deal with stressful situations either at work or in the family, or are suffering from stress for an extended period of time. Even children will sometimes say that they feel 'stressed'. A detailed discussion of the reasons for the rise in the perceived levels of stress nowadays is beyond the scope of this chapter, but the increased speed of life in general and the concurrent floods of information to be processed may play a prominent role. Naturally, uncertainties regarding the economy, together with an unstable job and housing market, present major causes of stress in many people's lives.

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When combined with a sedentary lifestyle and unhealthy eating habits, stress coping mechanisms may fail and the damaging effects of chronic stress on physical and mental health will become evident.

In principal, the human body is well-equipped to deal with single stressful events, but when stress becomes chronic (for example, during care of ill or older relatives, prolonged financial problems, etc.) it may precipitate disease. It is now well accepted that stress – or maybe, more accurately phrased, mal-coping with stress – is involved in the etiology of certain psychiatric diseases such as major depression and anxiety disorders. In depressed patients, major aberrations have been found in the regulation of the stress hormone system, the hypothalamic–pituitary–adrenocortical (HPA) axis, under both baseline and challenged conditions (De Kloet *et al.*, 2005). Thus, major depression is, for instance, associated with a flattened circadian rhythm of cortisol and with impaired suppression in the dexamethasone suppression test – a neuroendocrine challenge test to assess negative feedback efficacy within the HPA axis via the glucocorticoid receptor (GR). Analyses of CSF samples and *post-mortem* tissues of depressed patients have provided evidence for alterations at the level of the corticotropin-releasing factor (CRF) system, which is one of the key regulators of the activity of the HPA axis and also plays an important role in stress-related behaviors such as fear and emotion (Nemeroff *et al.*, 1984, 1988; Raadsheer *et al.*, 1994, 1995; Arborelius *et al.*, 1999). Furthermore, modern molecular techniques have shown that alterations at the level of the genome – in genes involved in HPA axis regulation – may render individuals more susceptible to the development of a mood or anxiety disorder (Binder *et al.*, 2004a; Derijk *et al.*, 2006; van Rossum *et al.*, 2006).

As described in detail elsewhere in this volume, the neurotransmitter serotonin (5-HT) plays an important role in the etiology of mood and anxiety disorders (see also Maes and Meltzer, 1995; Mann, 1998; Ressler and Nemeroff, 2000). Many of the available pharmacological treatments for these illnesses involve manipulating the 5-HT system in one way or another (e.g., selective serotonin reuptake inhibitors (SSRIs) and buspirone, a 5-HT_{1A} receptor partial agonist). Importantly, it has been found that alterations in genes of the 5-HT system, such as polymorphism in the promoter region of the 5-HT transporter gene, can moderate the impact of stressful life events on depression (Caspi *et al.*, 2003). Thus, given the prominent role of both the HPA axis and 5-HT in stress-related psychiatric disease, the question arises as to whether and how stress influences the 5-HT system and, furthermore, whether stress exerts its effects on 5-HT via components of the stress hormone axis (i.e., involving glucocorticoids and neuropeptides of the CRF family). We endeavor to answer these questions in this chapter as follows. In the next section, we will first review evidence demonstrating the responsiveness of the rostral serotonergic raphe nuclei to stressful challenges; this in terms of c-Fos expression in 5-HT neurons and changes in the firing rate of these neurons. Next, the effects of stress on the synthesis of 5-HT and on the expression of the rate-limiting enzyme tryptophan hydroxylase (TPH) will be discussed. The subsequent section will be devoted to the effects of stress on extracellular levels of 5-HT – as assessed by *in vivo* microdialysis – in regions of the forebrain which are essential for the coordination of the various components of the stress response. Finally, the consequences of manipulating the HPA axis and the CRF system for serotonergic neurotransmission will be considered. The data discussed in this chapter underscore the concept that stress impacts on the 5-HT system in a stressor- and brain-region-dependent manner, involving, under certain circumstances, glucocorticoid hormones and CRF and related neuropeptides. Given that current pharmacological therapies for the treatment of stress-related mental health diseases are still unsatisfactory – especially because of profound side effects, delay in clinical efficacy and a significant proportion of treatment-resistant patients – the intricate interaction between stress and 5-HT should be taken as a stimulant to further investigate the role of and identify new drug targets in these systems in depression and anxiety disorders. As outlined towards the end of the chapter, it will furthermore be of great importance to study the impact of moderate exercise on the 5-HT system in more detail so as to increase our understanding of why exercise is beneficial in the treatment of mental health diseases, and why, putatively, it could be used for the prevention of the development of such illnesses.

Effects of stress on the raphe nuclei: c-Fos expression and neuronal firing

In this chapter we will focus on the rostral 5-HT system, with cell bodies located in the dorsal raphe nucleus (DRN), the caudal linear nucleus and the median raphe nucleus (MRN), and providing a dense innervation to the whole forebrain (for neuroanatomical details, see Chapter 1.3 of this volume).

The immediate early gene c-Fos is often used as a marker to identify neurons that are activated by various manipulations (Kovacs, 1998). Using this strategy, it has become increasingly clear that stress indeed activates neurons within the DRN and MRN. Increased expression of c-Fos mRNA has been demonstrated after exposure of animals to different psychological stress paradigms, such as inescapable tail-shock, elevated plus maze (a behavioral test for the assessment of anxiety levels in rodents), social defeat stress, immobilization stress, restraint and

fear conditioning (Pezzone *et al.*, 1993; Silveira *et al.*, 1993; Watanabe *et al.*, 1994; Cullinan *et al.*, 1995; Beck and Fibiger, 1995; Martinez *et al.*, 1998; Grahn *et al.*, 1999; Ishida *et al.*, 2002; Rioja *et al.*, 2006; Lehner *et al.*, 2008). Forcing rats to swim – a stressor with not only psychological but also physical aspects (see below) – also induces c-Fos mRNA expression in both the DRN and MRN (Cullinan *et al.*, 1995). In contrast, activation of the immune system, which is regarded as a 'systemic' stressor representing an immediate physiological threat to the organism, seems to have no clear effect on c-Fos expression in the DRN and MRN in the rat (Ericsson *et al.*, 1994; Elmquist *et al.*, 1996; Laflamme *et al.*, 1999). However, in a recent study Hollis and colleagues demonstrated Fos-immunoreactivity in a specific subset of serotonergic neurons in the DRN (i.e., within the interfascicular and ventrolateral division of this raphe nucleus) after administration of bacterial endotoxin to mice (Hollis *et al.*, 2006). Interestingly, a topographically specific activation of neurons in the DRN has also been described after forced swim stress. Fifteen minutes of forced swimming in water at 25°C results in Fos-immunoreactivity in the dorsolateral DRN of rats (Roche *et al.*, 2003). Most of these Fos-positive neurons are, however, not 5-HT neurons but GABAergic interneurons, leading to the hypothesis that stress may regulate the neuronal activity of raphe 5-HT neurons (also) via its actions on the inhibitory neurotransmitter GABA within this nucleus. Given the observation that these Fos-positive neurons are enveloped by CRF fibers and contain CRF receptors (Roche *et al.*, 2003), it is of interest to note that intracerebroventricular administration of the CRF receptor type 1 (CRF1) ligands, CRF and urocortin 1, also causes a pronounced increase in Fos-immunoreactivity in the DRN (together with a moderate increase in the MRN) (Bittencourt and Sawchenko, 2000). In contrast, the CRF receptor type 2 (CRF2) ligand urocortin 2 has been found either to have no effect (Reyes *et al.*, 2001) or to increase the number of c-Fos-positive neurons mainly in the caudal regions of the DRN (Staub *et al.*, 2005). Administration of urocortin 2 directly into the caudal DRN results in an increased number of dual-labeled c-Fos/5-HT neurons in the caudal, middle and rostral DRN (Amat *et al.*, 2004). Interestingly, a recent study shows that in the caudal part of the DRN, levels of c-Fos expression correlate with the level of CRF-induced enhancement of the startle response in rats (Meloni *et al.*, 2008).

The above-described studies clearly show that stress (and members of the CRF neuropeptide family) results in a distinct and topographically specific neuronal activation of serotonergic (and other) neurons in the raphe nuclei. However, does this neuronal activation lead to changes in neuronal firing of 5-HT cell bodies in these nuclei? Unfortunately, data from studies aiming to answer this important question are so far inconclusive. Studies by Barry Jacobs and colleagues have shown, using an *in vivo* electrophysiology approach, that in the cat the firing pattern of 5-HT cell bodies remains unchanged (as compared to the firing rate during active waking) during various forms of stress, such as noise stress, pain, and thermoregulatory and glucoregulatory challenges (for references, see Jacobs and Azmitia, 1992). Also, predator exposure, a challenge which increases extracellular 5-HT levels in higher brain structures in rodents, was found to be without effect in the DRN of the cat (Wilkinson and Jacobs, 1988). In contrast, using a brain slice approach, Laaris and colleagues showed that a longer period of stress (30-minute restraint followed by 24-hour isolation or 16-hour exposure to novel uncontrolled environmental conditions), but not acute restraint stress, results in a decreased 5-HT_{1A} receptor-mediated inhibition of 5-HT cell firing rate in rats, which is dependent on corticosterone (Laaris *et al.*, 1997, 1999). Interestingly, early maternal separation (6 hours, 10 occasions, between postnatal days 5 and 20) has been found, *in vitro*, to decrease the sensitivity of DRN 5-HT neurons for α1-mediated excitation, but was without significant effect on the sensitivity of 5-HT_{1A} receptors. Furthermore, this manipulation did not change the proportion of classical and burst-firing 5-HT neurons, although the range of firing rates seems to be increased in the early-life stressed group (Gartside *et al.*, 2003). Subsequent work by the same group elegantly demonstrated that the balance between α1-mediated excitation and 5-HT_{1A}-mediated inhibition of the firing rate of DRN neurons is dependent on the level of corticosterone (Judge *et al.*, 2004). Thus, ADX rats implanted with a 2% corticosterone pellet show an enhanced α1-mediated excitation and a lower 5-HT_{1A}-mediated inhibition of DRN firing as compared to animals with a 70% corticosterone pellet (Judge *et al.*, 2004). By contrast, using an *in vivo* electrophysiology approach, it was found that prolonged (180-minute, postnatal days 2–14) maternal deprivation results in an increased 5-HT_{1A} autoreceptor-induced inhibition of raphe neuronal firing (Arborelius *et al.*, 2004). Studies on the effects of CRF on the firing rate of 5-HT neurons in the raphe nuclei have also provided evidence for a topographically specific influence. Thus, low doses of CRF were found to decrease the *in vivo* firing rate of 5-HT neurons in the rostral and medial parts of the DRN. Higher doses were without effect (Price *et al.*, 1998; Kirby *et al.*, 2000), whereas CRF increases the *in vitro* firing rate of a selective population of 5-HT neurons in the ventral and interfascicular region of the caudal DRN (Lowry *et al.*, 2000).

From the above, the picture is emerging that stress impacts on the activity of 5-HT neurons in the raphe

nuclei, but that it probably does so in a stressor- and topographically-specific way. Clearly more research is needed, implementing other stressful (acute and chronic) challenges and the effects of stress on non-classical (i.e., burst-firing) 5-HT neurons. Also, the role of glucocorticoid hormones needs to be further elucidated.

Effect of stress on the synthesis of 5-HT

5-HT is synthesized from L-tryptophan, an essential amino acid which needs to be taken up via the diet and is transported into the brain via an L-type neutral amino acid transporter in the blood–brain barrier. The rate-limiting step in the synthesis of 5-HT is the conversion of L-tryptophan into 5-hydroxytryptophan (5-HTP), a step which is catalyzed by the enzyme tryptophan hydroxylase (TPH). TPH is present in enterochromaffin cells, myenteric neurons of the gut, the pineal gland, and the brain raphe nuclei. Next, 5-HTP is converted into 5-HT via decarboxylation by the enzyme aromatic L-amino acid decarboxylase (AADC). There is some evidence that, under certain circumstances, the availability of L-tryptophan to the brain can affect the level of synthesis of 5-HT (Fernstrom and Wurtman, 1971; Knott and Curzon, 1972; Boadle-Biber, 1993).

TPH is activated when stores of 5-HT need to be refilled, as evidenced by the rise in the synthesis of 5-HT from L-tryptophan during electrical stimulation of serotonergic cell bodies (Shields and Eccleston, 1972; Herr *et al.*, 1975; Boadle-Biber *et al.*, 1986). As discussed above, stress has distinct effects on neuronal activation and, putatively, consequently on the firing rate of 5-HT neurons in the raphe nuclei, leading to the obvious question of whether stress also influences TPH activity and expression. After the initial study of Azmitia and McEwen describing that prolonged cold and electric foot-shock stress increased TPH activity in the midbrain and forebrain of rats (Azmitia and McEwen, 1974), Margaret Boadle-Biber and colleagues have performed a large series of studies characterizing the effects of loud sound stress on 5-HT synthesis using an *ex vivo* TPH activity assay in rats. Applying this assay, it was found that one session of randomly presented sound stress causes a short-lasting activation of TPH in the cortex. This increase in TPH activity is persistent even 24 hours later when rats are exposed to repeated sound stress sessions (Boadle-Biber *et al.*, 1989). Importantly, as pointed out above, the enhanced TPH activity is related to increases in neuronal firing rate, as no effects were found when 5-HT neurons were silenced by pretreatment of the animals with the 5-HT_{1A} receptor agonist gepirone (Corley *et al.*, 1992). The effects of stress seem to be dependent on the availability of glucocorticoids, as adrenalectomy has been found to block the rise in TPH activity caused by sound stress (Singh *et al.*, 1990). The CRF system may also play a role in the regulation of TPH activity. Intracerebroventricular administration of CRF increases TPH activity, an effect in which glucocorticoid receptors play a permissive role (Singh *et al.*, 1992).

From the above studies, it has become clear that stress indeed impacts on the activity of TPH. However, the next question that arises is whether stress affects 5-HT synthesis in a neuroanatomically specific manner. Using an assay measuring the accumulation of 5-HTP after inhibition of AADC by m-hydroxylbenzylamine, it was found that sound stress, forced swimming and tail-shock increased TPH activity in the MRN only, and were without effect in the DRN (Dilts and Boadle-Biber, 1995; Corley *et al.*, 2002). Furthermore, whereas CRF was found to increase THP activity in the MRN (Singh *et al.*, 1992), this neuropeptide had no effect in the mediobasal hypothalamus (Van Loon *et al.*, 1982). A neuroanatomically specific effect of stress is further underscored by studies utilizing immobilization stress. Immobilization stress had no effect on TPH activity in the hypothalamus, hippocampus, amygdala and DRN after five sessions (Palkovits *et al.*, 1976), although it seemed to increase TPH activity in the locus coeruleus and to decrease activity in the suprachiasmatic nucleus after seven sessions (Culman *et al.*, 1984). Furthermore, using an *in vivo* microdialysis approach to measure the accumulation of 5-HTP, it was found that 1-hour immobilization results in increased 5-HT synthesis in the nucleus accumbens and the medial prefrontal cortex of the rat (Nakahara and Nakamura, 1999).

So far, we have reviewed the literature regarding the effects of stress on the activity of the TPH enzyme. However, much recent research has focused on the expression of TPH mRNA and on the levels of the TPH protein. In 2003, Walther and colleagues, in a seminal study published in *Science*, described for the first time the existence of a second TPH gene, *Tph2* (Walther *et al.*, 2003). The previously known gene is now called *Tph1*. Importantly, they demonstrated that *Tph2* is expressed exclusively in the brain (Walther *et al.*, 2003), and it was subsequently found that TPH2 controls the synthesis of 5-HT in the brain (Zhang *et al.*, 2004). The highest expression of TPH2 mRNA is found in the ventromedial DRN and MRN at the midrostral level (Clark *et al.*, 2006). Interestingly, TPH2 mRNA levels follow a circadian rhythm which is under the control of the suprachiasmatic nucleus (SCN) and is mediated by the circadian rhythm of corticosterone (Malek *et al.*, 2007). In contrast to TPH2, TPH1 mRNA is abundantly expressed in the pineal gland and the periphery. Although some studies have described

the absence of *Tph1* in the raphe nuclei (Walther *et al.*, 2003; Cote *et al.*, 2003), the picture is now emerging that low levels of TPH1 mRNA are expressed in the brain serotonergic cell bodies (Patel *et al.*, 2004; Abumaria *et al.*, 2008) in rodents. In *post-mortem* human brain, low levels of TPH1 mRNA are found in the raphe nuclei but more substantial levels have been observed in several forebrain regions (e.g., the hypothalamus and amygdala) (Zill *et al.*, 2007). Furthermore, TPH1 may play a role in the late stage of development, as peak levels of TPH1 mRNA at P21 have been described recently (Nakamura *et al.*, 2006). The effects of stress on the expression of the two *Tph* genes have not been fully clarified yet. Although TPH2 seems to be the most important enzyme form for 5-HT synthesis in the brain, no effects of 1-week daily restraint stress on TPH2 mRNA levels were found (Abumaria *et al.*, 2008). Also, chronic social stress was found to have no effect on the expression of TPH2 mRNA (Abumaria *et al.*, 2006). Furthermore, stress during early gestation seems to only cause marginal (trend to significance) effects on TPH2 mRNA in the DRN of male offspring (Mueller and Bale, 2008), whereas chronic administration of ovine CRF or urocortin 2 had no effect on the absolute expression of TPH2 mRNA in the DRN (although changes in the relative expression between the midrostral ventromedial and core dorsomedial DRN were observed) (Clark *et al.*, 2007). Recently it has been described that daily subcutaneous administration of the glucocorticoid receptor agonist dexamethasone (0.3–3 mg/kg subcutaneously for 4 days) decreased TPH2 protein levels in the raphe nuclei of mice, but, given the poor penetration of dexamethasone into the brain, the underlying mechanism of this observation needs to be further elucidated (Clark *et al.*, 2008). With respect to the marginal effects of stress on TPH2 described so far, it is interesting that, despite its low levels of expression in the brain, TPH1 is more clearly affected by stress. Thus, repeated immobilization or restraint stress has been found to substantially (by 2- to 10-fold) increase TPH1 mRNA levels in the DRN and MRN (Chamas *et al.*, 1999, 2004; Abumaria *et al.*, 2008). However, the effect of repeated immobilization stress on TPH1 mRNA expression seems to be independent of circulating glucocorticoids, as similar effects of the stressor were observed in adrenalectomized rats (Chamas *et al.*, 2004). The above-described findings clearly demonstrate the need for further research into the effects of stress on the two forms of TPH (at the gene and protein level), taking into account the possible stressor dependency and neuroanatomical specificity of such effects. Such studies will be of great importance not only for our basic understanding of the regulation of serotonergic neurotransmission, but also, given the changes observed in *Tph2* expression in suicide victims (Bach-Mizrachi *et al.*, 2008) and the putative relationship between *Tph1* and *Tph2* gene variants and major depression (Zill *et al.*, 2004; Gizatullin *et al.*, 2006; Galfalvy *et al.*, 2008, but also see Gizatullin *et al.*, 2008), for further clarifying the role of 5-HT in stress-related psychiatric illnesses.

Effects of stress on the extracellular levels of 5-HT and 5-hydroxyindoleacetic acid

During the past 25 years, *in vivo* microdialysis has become the method of choice to study extracellular levels of neurotransmitters in the brain. Microdialysis is predominantly used in neuropharmacology to evaluate the effects of (newly developed) drugs on neurotransmitter systems, and to elucidate their mechanisms of action. However, microdialysis is increasingly becoming an important tool to study the role of neurotransmitter systems in physiology and behavior. As we have recently discussed (Linthorst and Reul, 2008), several refinements in the performance of the technique, including the use of gas anesthesia, guide cannulas and special freely-moving animal systems, together with improved analytical methods, have made microdialysis extremely suitable for the study of various aspects of the stress response in rats and (mutant) mice (Figure 1). Below we will discuss the effects of stress on the extracellular levels of 5-HT and its metabolite 5-hydroxyindoleacetic acid (5-HIAA) in (1) three higher brain regions important in the behavioral and psychological aspects of the stress responses (i.e., the hippocampus, the frontal cortex and the amygdala) and (2) the hypothalamus and pre-optic area, because of their key role in the regulation of the endocrine and autonomic responses to stress.

Hippocampus, prefrontal cortex and amygdala

During recent years it has become clear that stress affects hippocampal levels of 5-HT and its metabolite 5-HIAA in a stressor-dependent way. Submitting rodents to predator stress, an ethologically highly relevant psychological stressor, has been found to increase extracellular levels of 5-HT and 5-HIAA in the hippocampus of rats and mice (Rueter and Jacobs, 1996; Linthorst *et al.*, 2000). In a recent study, we used a rapid sampling design (5 minutes during and immediately after the stressful challenge) to study the effects of exposure to a rat on the serotonergic neurotransmission in the hippocampus of male C57BL/6N mice (Beekman *et al.*, 2005). For this stressor, a male Wistar rat was introduced into the home cage of

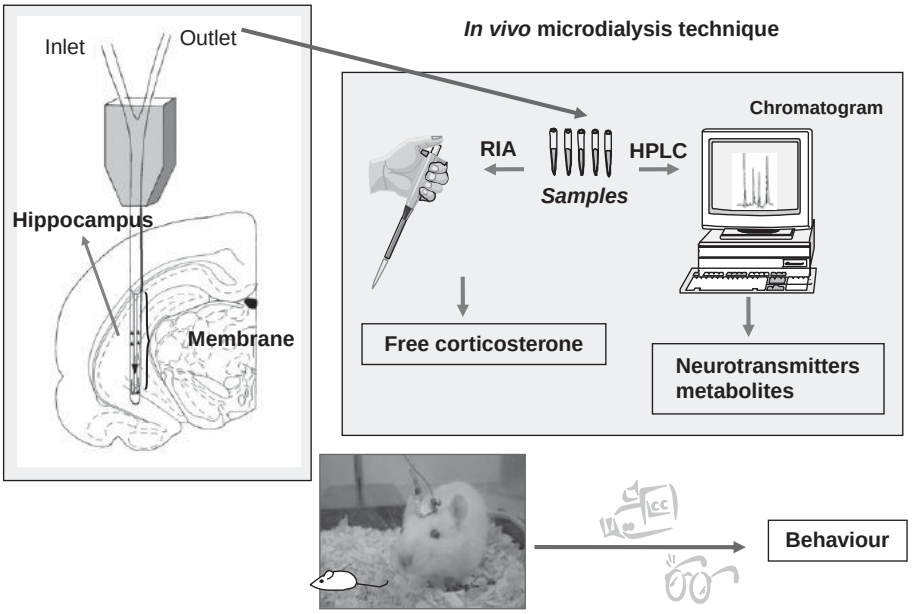

Figure 1 Schematic overview of the microdialysis technique as applied in our laboratories. A microdialysis probe is inserted through a guide cannula (not depicted, implanted 7–10 days earlier; the dialysis probe depicted is the CMA12 of CMA Microdialysis/AB, Stockholm, Sweden) exposing the dialysis membrane to the brain tissue of interest (shown here is the hippocampus). The microdialysis probe is perfused with Ringer solution via a microinfusion pump (not depicted). Drugs can be delivered to the target tissue by inclusion in the perfusion fluid (so-called retrodialysis or reverse dialysis). Animals are connected to a swivel system and counterbalance arm via a little peg anchored to the dental cement on their skull (see picture) giving them the opportunity to move freely in all directions. Samples are collected using automated microsamplers, and are subsequently used to measure various molecules in the dialysates – e.g., neurotransmitters and their metabolites – by high-pressure liquid chromatography (HPLC) with electrochemical detection or free corticosterone levels via radioimmunoassay (RIA). Behavior of the rats or mice is continuously scored during the microdialysis experiments by visual observation or is subsequently analyzed from videotape. Experiments are performed under baseline conditions, and during and following stressful challenges or behavioral testing. Note that this method allows the simultaneous collection of information on the status and responsivity of various neurotransmitter systems in the brain, on the activity of the HPA axis and on behavior, providing strong, integrative data and representing a clear example of refinement and reduction (Russell and Burch, 1959) in animal research. To see the full color version of this figure, please refer to the color plate in the back of the book. Copies produced via our print on demand service do not contain color plates. If your copy does not have the color plate, please go to this website to view the figure in color www.elsevierdirect.com/companions/9780123746344

the mouse but remained separated from the mouse by a Plexiglas wall with small holes. Even during the first 5 minutes of the total 30 minutes of rat exposure, 5-HT levels increased rapidly, reaching maximum levels of about 300 percent of baseline. 5-HT remained elevated during the next 10 minutes, but started to decrease in the second half of the rat exposure procedure. The levels of 5-HIAA also increased with the introduction of the rat, but this response was somewhat delayed as compared to the immediate 5-HT response. It is of interest to contrast the effects of the presence of the predator on hippocampal 5-HT with the behavioral changes concurrently taking place in the stressed mice. Immediately at the start of exposure to the rat, mice usually move to a corner of the cage and become highly vigilant. Next, various risk-assessment behaviors are initiated, such as sniffing the air, stretching towards and retracting from the separation wall (flat back posture) and then sniffing at the separation wall. During the second half of the predator exposure, mice were seen to be extensively sniffing the bedding of their cage (Beekman *et al.*, 2005). From these results we concluded that during psychological stress 5-HT in the hippocampus increases especially during the initial phase, supporting the assessment of a new and potentially life-threatening situation and assuring adequate coping responses regarding the challenge. Increases in extracellular concentrations in 5-HT and 5-HIAA were also observed in the frontal cortex, the amygdala and lateral septum (Rueter and Jacobs, 1996; Beekman *et al.*, 2005).

Whereas the above-discussed predator stress paradigm is an example of a psychological form of stress, most stressors used to investigate the effect of stress on hippocampal 5-HT have combined psychological and physical aspects. Tail pinch (attaching a paper clip to the tail of a rat or mouse) causes arousal and also activates pain sensory pathways and consummatory behaviors. Microdialysis studies have shown that this form of stress results in increases in extracellular 5-HT in the

hippocampus, prefrontal cortex and amygdala, although in some studies this response was delayed until after termination of the stressor (Kalén *et al.*, 1989; Pei *et al.*, 1990; Vahabzadeh and Fillenz, 1994; Rueter and Jacobs, 1996; Fujino *et al.*, 2002). The delayed increase in 5-HT could well be caused by the increased consummatory and grooming behavior after the removal of the paper clip – a notion that may be underscored by the demonstration of a subset of 5-HT neurons which is highly active during oral–buccal movements in the cat (Fornal *et al.*, 1996), and by our own observation that grooming behavior in the home cage profoundly increases hippocampal 5-HT in rodents (Linthorst *et al.*, unpublished data). Electric foot- or tail-shock represents another example of a stressor with combined physical and psychological aspects. Surprisingly, there is not much information available regarding the effects of this stressor on extracellular levels of 5-HT in forebrain regions. *Post-mortem* tissue studies suggest that electrical foot-shock increases 5-HT turnover in the hippocampus, amygdala and prefrontal cortex (for references, see Linthorst, 2005). Interestingly, it seems that the exact response to electrical shock is dependent on the extent of control an animal can exert over the shock. Thus, an augmented turnover and extracellular response of 5-HT has been found in the frontal cortex and ventral hippocampus of rats for whom the shock was inescapable, as compared to their conspecifics that could escape the shock (Amat *et al.*, 1998; Heinsbroek *et al.*, 1991). Interestingly, the prefrontal cortex plays a key role in inhibiting the activation of serotonergic neurons in the DRN when rats are given the opportunity to escape shocks – i.e., when the stressor is controllable by the animal (Amat *et al.*, 2005). Conditioned fear stress also leads to an increase in hippocampal and prefrontal cortical extracellular levels of 5-HT, although the neurotransmitter response was not studied during the training phase when animals actually receive the electrical shocks (Yoshioka *et al.*, 1995; Wilkinson *et al.*, 1996; Hashimoto *et al.*, 1999). Importantly, Wilkinson and colleagues demonstrated that the 5-HT activation in the hippocampus induced by conditioned fear stress was due to the contextual aversive cues (i.e., the box in which the animal had received the shocks) but not to the conditioned discrete stimulus (Wilkinson *et al.*, 1996).

During the past years, we and others have extensively studied the effects of the combined psychological/physical stressor forced swimming on extracellular levels of 5-HT and 5-HIAA. Forced swim stress is performed by placing rats or mice in a glass beaker filled with water from which they cannot escape. Forced swim stress causes profound behavioral changes. During the swim session, animals will first try to escape by climbing against the wall and, in the case of rats, will explore possible escape routes under the water surface (diving). This period is followed by swimming in the container and by floating behavior. Although increased floating behavior/immobility is used as an index for despair or depressed behavior in the Porsolt swim test, it is now increasingly seen as an index of a learning process – i.e., the animal learns a behavioral coping strategy aimed to conserve energy when forced to swim on a second exposure (De Pablo *et al.*, 1989; Abel, 1993; Korte, 2001; Bilang-Bleuel *et al.*, 2005; Chandramohan *et al.*, 2008). However, apart from increased motor activity during parts of this challenge, the physiology of the animals is compromised by a rapid and extensive loss of body temperature (Stone, 1970; Linthorst *et al.*, 2007) (see below). Fifteen minutes of forced swim stress at 25°C results in an immediate increase in hippocampal levels of 5-HT. This increase is followed by a return to pre-stress levels when animals are returned to their home cage, and a subsequent second rise in the neurotransmitter levels (Figure 2C). Interestingly, whereas swimming in warm water (35°C) causes a rapid and prolonged rise in hippocampal extracellular 5-HT (Figure 2E), no effect on 5-HT was found when the animals were forced to swim in cold water (19°C), although levels increased 15–30 minutes after the termination of the stressor (Figure 2A). The effects of forced swim stress on extracellular levels of 5-HIAA were also highly water-temperature-dependent. Thus, forced swim stress in water at 35°C causes an immediate and sustained increase in hippocampal extracellular 5-HIAA (Figure 2F). However, the levels of this metabolite actually first decrease significantly in rats forced to swim in cold water at 19°C, after which they increase gradually (Figure 2B) (Linthorst *et al.*, 2007). Using an *in vivo* biotelemetry approach, it was found that 15 minutes of forced swimming had minor effects on the body temperature of the rats when water of 35°C was used (a decrease of about 1°C). However, at 25°C and 19°C, rats lost, respectively, approximately 8°C and 12°C of body temperature (Figure 2) (Linthorst *et al.*, 2007). These observations, together with the microdialysis findings described above, have led us to postulate that the final outcome of the neurotransmitter response to stress is shaped by the combination of the psychological aspects of the stressor and the impact of the concurrent physical changes. Recent investigations into the effects of forced swim stress on extracellular levels of GABA in the hippocampus have further substantiated this hypothesis. Thus, forced swim stress at 35°C increased hippocampal levels of GABA, whereas swimming at 25°C led to a profound decrease in extracellular neurotransmitter concentrations in this brain structure (De Groote and Linthorst, 2007). The physical impact of forced swim stress may also explain why there are some conflicting results reported in the literature with regard to the effects of this stressor on hippocampal 5-HT. Thus, an

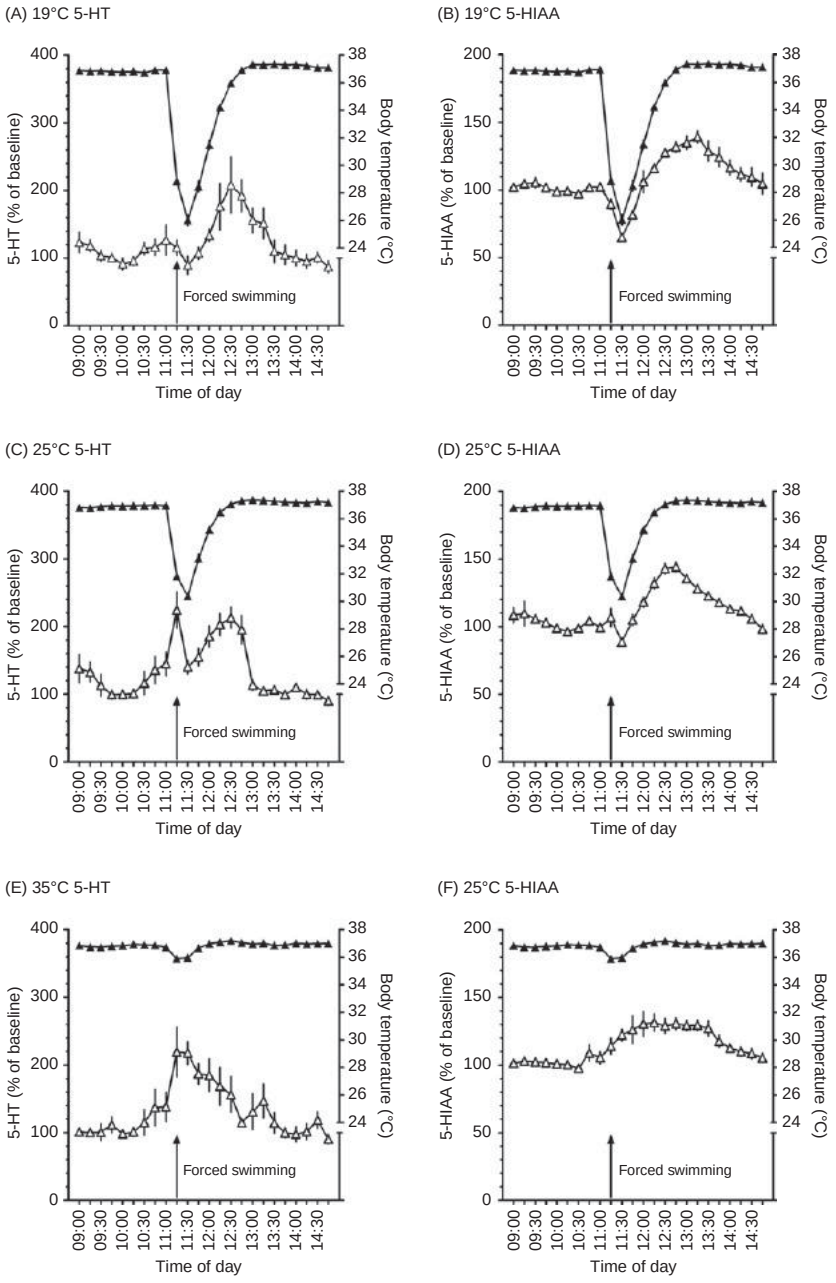

Figure 2 Comparison of the effects of forced swim stress at different water temperatures – (A, B) 19°C; (C, D) 25°C; (E, F) 35°C – on core body temperature (closed triangles) and hippocampal extracellular levels of 5-HT and 5-HIAA (open triangles) in male Wistar rats. 5-HT and 5-HIAA levels were assessed by *in vivo* microdialysis and body temperature (in separate animals) by *in vivo* biotelemetry. Levels of 5-HT and 5-HIAA are expressed as percentage of baseline. The time points on the x-axis indicate the time of the day at which collection of the respective sample was started. After collection of nine 15-min baseline samples, rats were placed in a beaker glass with water for 15 min (at 11:15 h, indicated by the arrow). After the forced swim session they were taken out of the water, dried with a towel and returned to their home cage, following which another fourteen 15-min dialysis samples were collected. 5-HT and 5-HIAA concentrations were analyzed by HPLC with electrochemical detection. Body temperature data were collected at 2-min intervals, but were averaged over 15-min bins to match the 15-min dialysate sample duration. Values are mean ± SEM. For 5-HT and 5-HIAA $n = 5$–7; for body temperature $n = 6$. Note that swimming for 15 min at 19 and 25°C produced a pronounced decrease in core body temperature and that the 5-HT and 5-HIAA responses in the hippocampus are dependent on the water temperature used. Comparison of the body temperature and neurotransmitter profiles suggests that two different processes may shape the ultimate 5-HT and 5-HIAA responses to swim stress. Thus, forced swimming causes an increase in extracellular levels of 5-HT and 5-HIAA, which is temporarily quenched or concentrations are transiently decreased when core body temperature falls below approximately 31°C. Reproduced from Linthorst *et al.* (2007), with permission of *Stress* (Informa Healthcare).

increase in hippocampal 5-HT was found by Rueter and Jacobs in rats that were forced to swim for 30 minutes in water of 30–35°C (Rueter and Jacobs, 1996), while Kirby and colleagues only observed a decrease in extracellular 5-HIAA (and no effect on 5-HT) when rats had to swim for 30 minutes in water of 21–22°C (Kirby *et al.*, 1995). Moreover, these authors found also disparate effects of swim stress in the frontal cortex and amygdala – i.e., increased levels or no effect/decreased levels of 5-HT and 5-HIAA (Kirby *et al.*, 1995; Rueter and Jacobs, 1996).

Immune stress is an example of a stressor with an immediate threat to the integrity and survival of an organism, and therefore is expected to merely be processed at the level of lower brain structures such as the hypothalamus and brainstem nuclei (Herman and Cullinan, 1997). However, we have provided extensive evidence that an immunological challenge impacts on serotonergic neurotransmission in the hippocampus of rats. Thus, systemic administration of bacterial endotoxin (lipopolysaccharide, LPS) results in a pronounced and prolonged (>6 hours) elevation of hippocampal extracellular levels of 5-HT and 5-HIAA (Linthorst *et al.*, 1995a). These effects can be mimicked by peripheral and intracerebroventricular administration of interleukin-1β and interleukin-2, but not by TNF-α (Linthorst *et al.*, 1995a; Merali *et al.*, 1997; Pauli *et al.*, 1998). Studies in rats treated chronically with CRF suggest that a full hippocampal 5-HT response is a prerequisite for the adequate development of sickness behavior during immune activation (Linthorst *et al.*, 1997). Based on these observations, we have concluded that immune stress also possesses a distinct psychological component probably involved in the mood changes occurring during infection.

Hypothalamus and pre-optic area

The hypothalamus and pre-optic area are key structures in the coordination of the HPA axis and the autonomic response to stress. Most studies using *post-mortem* tissues have demonstrated that stress increases or has no effect on the turnover of 5-HT in the different subnuclei of the hypothalamus (for references, see Linthorst, 2005). Surprisingly, there is little information available regarding the effects of stress on extracellular levels of 5-HT and 5-HIAA as assessed by *in vivo* microdialysis. We have studied the effects of immune stress on serotonergic neurotransmission in the pre-optic area of the rat. In contrast to the profound activation of 5-HT in the hippocampus as described above, no effect on extracellular 5-HT and only a minute rise in extracellular 5-HIAA was found in the pre-optic area (Linthorst *et al.*, 1995b). The absence of an effect of immune stress on pre-optic 5-HT is underscored by similar observations made in the cat (Wilkinson *et al.*, 1991). In contrast, immune system activation was found to increase both extracellular 5-HT and 5-HIAA levels in the hypothalamus of the rat (Mohankumar *et al.*, 1993; Shintani *et al.*, 1993; Lavicky and Dunn, 1995). With respect to psychological stress, we have recently demonstrated that exposure to a predator increases extracellular levels of 5-HT and its metabolite in the paraventricular nucleus and anterior hypothalamus of Balb/c mice (Beekman and Linthorst, unpublished observations). Another form of psychological stress, immobilization, was also found to increase markedly the extracellular levels of 5-HT in the hypothalamus (Shimizu *et al.*, 1992; Shintani *et al.*, 1995), although only a marginal effect on 5-HIAA was found. Interestingly, these effects could be blocked by infusion of an IL-1 receptor antagonist (Shintani *et al.*, 1995). From the above, the general picture is emerging that stress increases extracellular 5-HT in the hypothalamus (but not the pre-optic area), while it is at the same time clear that more research (implementing both acute and chronic stress paradigms) will be needed to further substantiate this notion.

Repeated and chronic stress

In previous sections of this chapter we have given examples of studies demonstrating that repeated stress and stressors of a longer duration may change the electrophysiological properties of raphe 5-HT neurons and the synthesis of this neurotransmitter. However, similarly to 5-HT firing rate and synthesis, there is, unfortunately, only limited information available regarding the effects of repeated or chronic stress on extracellular levels of 5-HT and 5-HIAA. This may have to do with the complicating factor that microdialysis is normally performed over a restricted number of days. However, there are a few interesting studies which have investigated the status and responsiveness of the 5-HT system at the end of a chronic stress paradigm.

Under baseline conditions, extracellular levels of 5-HT are closely correlated with the vigilance state of the animal. Thus, in various brain structures, among them the hippocampus, highest levels of 5-HT are found during active waking, while levels decrease when animals are entering slow-wave sleep (Houdouin *et al.*, 1991; Imeri *et al.*, 1994; Portas and McCarley, 1994; Portas *et al.*, 1998; Park *et al.*, 1999; Shouse *et al.*, 2000; Python *et al.*, 2001; Peñalva *et al.*, 2003). Interestingly, extracellular levels of 5-HT are lowest during rapid eye movement sleep, probably caused by the almost complete silencing of the firing rate of raphe 5-HT neurons during this sleep stage. In a combined *in vivo* microdialysis/sleep EEG study, we found that during a longer period of stress

(i.e., a 4-hour period of sleep deprivation), hippocampal 5-HT levels are increased throughout the whole period (Peñalva *et al.*, 2003). However, after termination of the sleep-deprivation procedure, hippocampal levels of 5-HT decrease during the rebound phase and the normal 5-HT/sleep stage relationship is immediately re-established. Of interest in this respect is a recent study demonstrating that a chronic mild stress paradigm is able to disrupt exactly this tight relationship between hippocampal 5-HT levels and sleep stage by an as yet unknown mechanism (Gronli *et al.*, 2007).

Another example of the effects of chronic stress on 5-HT levels is the isolation-rearing model, as developed by Charles Marsden. These authors have studied the behavioral and neurochemical consequences of isolation rearing – i.e., single housing of rats for 6 weeks immediately following weaning – as compared to social, group housing. Isolation induces profound behavioral changes in the animals, but no changes in baseline 5-HT levels in the dorsal hippocampus were observed. However, the 5-HT system is clearly compromised in isolation-reared rats when challenged with a stress paradigm. Thus, foot-shock stress and stress caused by placing the animal on an elevated x-maze both result in a rise in hippocampal 5-HT in group-reared rats, but have no effect in isolated rats (Bickerdike *et al.*, 1993; Muchimapura *et al.*, 2002). There are some indications that the effects of chronic stress on extracellular 5-HT may be stressor-dependent. Repeated exposure of rats to an elevated open platform not only increased extracellular levels of 5-HT and 5-HIAA in the dorsal hippocampus under baseline conditions, but also resulted in elevated 5-HT levels when animals were tested on day 10 of the repeated platform exposure; this as compared to naïve animals which showed no changes in hippocampal 5-HT when placed on the elevated open platform for the first time (Storey *et al.*, 2006).

Effects of stress on serotonergic neurotransmission in animal models of aberrant HPA axis functioning

As discussed above, changes at the level of the genome may be implicated in the etiology of stress-related psychiatric disease. Mutant mouse strategies may therefore represent a useful tool to further study the regulation of the stress response and, importantly, to gain insight into (malfunctioning) stress coping mechanisms. Based on the putative involvement of the CRF system in depression and anxiety, and our observation that long-term elevation of brain CRF levels precipitate altered responsiveness of hippocampal 5-HT to stress (Linthorst *et al.*, 1997), we became interested in studying a genetic mouse model of CRF1 impairment (Timpl *et al.*, 1998). Performance of *in vivo* microdialysis in freely behaving CRF1 knockout mice over the diurnal cycle demonstrated that deletion of CRF1 results in significantly elevated levels of 5-HIAA, but not of 5-HT, in the hippocampus during both the light and dark periods (Peñalva *et al.*, 2002). There is substantial evidence that 5-HIAA levels may be an indicator for newly-synthesized 5-HT which has not been released from the presynaptic element (Grahame-Smith, 1974; Kuhn *et al.*, 1986; Peñalva *et al.*, 2002). Thus, the increased levels of 5-HIAA suggest that the synthesis of 5-HT is increased (in the hippocampus) in CRF1 knockout mice. Naturally, the question arises whether the deletion of CRF1 also has consequences for the response of serotonergic neurotransmission to stress. To address this question, CRF1 knockout and wild-type mice were subjected to a 10-minute session of forced swim stress (25°C). Interestingly, although the 5-HIAA response is similar in both genotypes, CRF1 knockout mice show an augmented 5-HT increase in the hippocampus when subjected to this form of stress (Peñalva *et al.*, 2002). To the best of our knowledge, this is the only information available on extracellular 5-HT and 5-HIAA in a mutant mouse model with alterations at the level of the CRF system. Together with the pronounced effects of CRF and the urocortins on hippocampal 5-HT and 5-HIAA levels demonstrated after intracerebroventricular administration (De Groote *et al.*, 2005), these data clearly show the intricate relationship between the CRF and 5-HT systems; aberrant functioning of this relationship may therefore underlie the development of certain psychiatric diseases.

Apart from changes in CRF, changes in GR and at the level of the mineralocorticoid receptor, both critical in the regulation of the HPA axis, may also play a role in stress-related mental health disorders. In an earlier study we investigated the stress responsiveness of GR-impaired mice with respect to behavior, HPA axis activity and hippocampal serotonergic neurotransmission (Linthorst *et al.*, 2000). Exposure of GR-impaired mice to a predator (rat; see above) resulted in an abnormal behavioral response to this potentially life-threatening stressor. GR-impaired mice showed abnormal risk assessment behaviors (i.e., an immediate inspection of the separation wall) which was accompanied by an absent HPA axis response to this challenge. Interestingly, during predator stress an augmented rise in 5-HT levels in the hippocampus in GR-impaired mice was found. Thus, although impairment of GR did not lead to changes in baseline levels of 5-HT and 5-HIAA, the response of this neurotransmitter to psychological stress was clearly exaggerated (Linthorst *et al.*, 2000). Alterations in serotonergic neurotransmission have also been found in other HPA axis-related mutant mice (Wei *et al.*, 2004; Rozeboom *et al.*, 2007), but unfortunately these studies were limited to the measurement of 5-HT_{1A}

receptor mRNA expression, and to date no studies on the extracellular levels of 5-HT and its metabolite in these mouse lines have been published.

Enhanced stress coping by exercise

As put forward in the introduction of this chapter, understanding how stress affects serotonergic neurotransmission is essential for increasing our insight into the mechanisms underlying stress-related psychiatric disease. However, equally important will be a further understanding of factors and mechanisms promoting adequate psychological and physiological coping with stress. There is now abundant evidence demonstrating that moderate exercise has beneficial effects on the cardiovascular system, on weight and food intake control and on sleep profiles (Lancel *et al.*, 2003). Moreover, during the past few years we have provided extensive evidence showing that voluntary exercise (i.e., wheel running in rats and mice that have access to a running wheel in their home cage) increases the flexibility of the HPA axis response to a stressful situation (Droste *et al.*, 2003, 2007). Thus, when corticosterone is needed to mobilize energy during a stressor with physical impact, such as during forced swimming, the plasma response of this stress hormone is markedly enhanced as compared to the response in sedentary animals. In contrast, during a mild psychological stressor, such as exposure to a novel environment, the plasma corticosterone response is significantly reduced in exercised rats and mice (Droste *et al.*, 2003, 2007). Interestingly, the lower plasma corticosterone response correlates with less anxious and risk-assessment behaviors during the period of exposure to novelty, suggesting a better behavioral and physiological coping with the stressful challenge in exercised animals (Droste *et al.*, 2007; Collins *et al.*, 2009). The notion that the dynamics of the HPA axis may be increased in exercising subjects is also underscored by an increased amplitude in the ultradian rhythm of free corticosterone (as assessed by *in vivo* microdialysis; see Droste *et al.*, 2008; Linthorst and Reul, 2008) in exercised rats as compared to sedentary control animals in the hours before the start of the active period (Droste et al., 2009). The above described evidence clearly supports a beneficial role of exercise in the regulation of physiological parameters, but, also for the purpose of this chapter, the logical next question would be whether exercise also supports mental health and, if so, which neurotransmitter pathways would be involved. Although exercise is increasingly being prescribed as a co-treatment for a number of psychiatric illnesses, its scientific underpinning needs to be strengthened. We have demonstrated that exercised mice show reduced anxiety and less impulsivity in a number of behavioral tests (Binder *et al.*, 2004b). Moreover, exercised rats show profound alterations in the expression of GABA-A receptor subunits in the brain (Hill *et al.*, 2008), which is of interest given that GABA-A receptors are an important drug target for the treatment of stress-related psychiatric disease. But what about 5-HT? Unfortunately, research on the effects of exercise on serotonergic neurotransmission in higher limbic brain regions has mainly been conducted using a forced running paradigm – i.e., forcing rodents to run on a treadmill. This paradigm causes stress in rodents, as it is usually performed during the light period – i.e., the period when the animals would normally rest and sleep. With this caveat in mind, treadmill experiments indicate that exercise may increase 5-HT and 5-HIAA levels in various brain structures (Meeusen *et al.*, 1996; Wilson and Marsden, 1996; Gomez-Merino *et al.*, 2001), which would fit with the observation that there is a close relationship between extracellular levels of 5-HT and vigilance state (with highest levels attained during active waking), as described above. Interestingly, recently increased expression of TPH2 mRNA in the DRN and MRN at zeitgeber time 10 (but not at zeitgeber time 2) has been described in rats after voluntary wheel running for 6 weeks (Malek *et al.*, 2007). Moreover, voluntary exercise significantly reduces the levels of 5-HT transporter mRNA in the DRN and MRN after 3 and 6 weeks (Greenwood *et al.*, 2005). However, from the limited available information, it is clear that, for a better understanding of the beneficial effects of exercise on mood, affect and anxiety, it will be of critical importance to further study the impact of exercise on the 5-HT system (and on other neurotransmitter systems including GABA) during voluntary exercise and using an *in vivo* microdialysis paradigm.

Concluding remarks and future

As reviewed in this chapter, the current state of our knowledge on the relationship between stress and the 5-HT system clearly shows that this neurotransmitter system is highly responsive to acute forms of stress, albeit in a stressor-dependent and brain region-specific manner. Without doubt, the available information on the consequences of chronic stress on serotonergic neurotransmission is limited, and no firm conclusions can be drawn. In our opinion it will be of great importance to further elucidate the effects of stress on the 5-HT system with more attention to the impact of chronic stress and to the mechanisms underlying putative alterations in serotonergic neurotransmission. Studies implementing the 5-HT system in exercising subjects will be needed to further increase our knowledge on how to improve stress coping

mechanisms, and will contribute to our understanding of how to increase stress resilience in people suffering from stress in today's hectic and stressful world.

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